

19-07-2019

Bachelor of Pharmacy II Semester- End Semester Examination- July - 2019
3621

HUMAN ANATOMY AND PHYSIOLOGY-II

Instructions to the candidates:

1. Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules.
2. Section-A contains Q-1 and Q-2 with Five (5) questions each. All the Ten very short type questions are compulsory and carry equal marks ($10 \times 2 = 20$).
3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. ($2 \times 10 = 20$).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Action potential
 - b) ATP
 - c) Lung volume
 - d) DNA
 - e) RAS
2. Answer the following questions in very short (Maximum 20 Words): 5 x 2 = 10
 - a) What is reflex activity?
 - b) What are meninges?
 - c) What do you mean by fertilization?
 - d) Write role of pancreas.
 - e) Give names of two disorders of GIT.

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. Enumerate different types of RNA. Explain process of protein synthesis with diagram.
4. Enlist various hormones of adrenal gland. Give functions of adrenal gland.
5. Discuss mechanism of respiration with suitable diagram.

Section – C

6. Answer (any Four) of the following questions in short answer: 4 x 5 = 20
 - a) Give physiological role of small intestine.
 - b) Explain properties of nerve fiber.
 - c) What do you understand by acid production in stomach? Explain.
 - d) Write anatomy of kidney.
 - e) Describe genetic pattern of inheritance.
7. Answer (any Three) of the following questions in short answer: 3 x 5 = 15
 - a) Write classification of hormones.
 - b) Describe disorders of kidney.
 - c) Discuss formation of ATP.
 - d) Give ventricles of brain.

PHARMACEUTICAL ORGANIC CHEMISTRY-I

Instructions to the candidates:

1. Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules.
2. Section-A contains Q-1 and Q-2 with Five (5) questions each. All the Ten very short type questions are compulsory and carry equal marks ($10 \times 2 = 20$).
3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. ($2 \times 10 = 20$).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 2

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Explain Huckel's rule for aromatic compounds.
 - b) Explain the term cracking.
 - c) What is vicinal dihalide compound?
 - d) Define Markovnikov's rule with suitable example.
 - e) Explain ozonolysis reaction of alkene.
2. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Explain allylic rearrangement of alkene.
 - b) Explain electromeric effect of carbonyl compounds.
 - c) Mention the structure and uses of chloral hydrate.
 - d) Define acidity of carboxylic acids.
 - e) Mention the structure and uses of Iodoform.

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. What are organic compounds? Classify them with suitable examples.
4. Explain E_1 and E_2 reactions. Differentiate E_1 versus E_2 reactions and also mention factors affecting E_1 and E_2 reactions.
5. Write method of preparation and chemical reaction (Any two) of alcohols. Also discuss qualitative test of alcohol.

Section – C

6. Answer (any Four) of the following questions in short answer:

4 x 5 = 20

- a) Define and classify isomerism.
- b) Explain Diel-Alder reaction with mechanism and example.

c) Explain Markownikoff's and anti-Markownikoff's rule.

d) Explain Aldol condensation reaction.

e) Explain Cannizzaro reaction.

7. Answer (any Three) of the following questions in short answer:

3 x 5 = 15

a) How will you distinguish primary, secondary and tertiary alcohols? Explain any one test.

b) Explain Benzoin condensation with mechanism.

c) Explain Perkin reaction with mechanism.

d) Enlist the factors affecting SN^1 and SN^2 reactions.

Instructions to the candidates:

1. Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules.
2. Section-A contains Q-1 and Q-2 with Five (5) questions each. All the Ten very short type questions are compulsory and carry equal marks ($10 \times 2 = 20$).
3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. ($2 \times 10 = 20$).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Give classification and chemical nature of carbohydrates.
 - b) Define energy rich compound and classify them.
 - c) Define glycolysis.
 - d) Discuss steps of citric acid cycle.
 - e) Discuss about mechanism of electro transport chain.
2. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Discuss about ketoacidosis.
 - b) What is Hypercholesterolemia?
 - c) What is jaundice and discuss its types?
 - d) Give structure and function of DNA.
 - e) Discuss nomenclature of enzymes.

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. Discuss synthesis and biological significance of 5HT.
4. Discuss biological role of proteins.
5. Discuss about hormonal regulation of diabetes mellitus and also comment on atherosclerosis.

Section – C

6. Answer (any Four) of the following questions in short answer: 4 x 5 = 20
 - a) Give classification and nature of lipids.
 - b) Discuss biological significance of ATP.
 - c) Discuss pathway, energetic and significance of glycolysis.
 - d) Discuss biosynthesis of fatty acids.
 - e) Discuss biosynthesis of pyrimidine nucleotides.
7. Answer (any Three) of the following questions in short answer: 3 x 5 = 15
 - a) What are enzymes and its classification?
 - b) Discuss about hyperuricemia, gout?
 - c) Discuss in brief about protein synthesis and inhibitors.
 - d) Discuss electron transport chain and its significance.

PATHOPHYSIOLOGY

Instructions to the candidates:

1. Do not write anything on question paper except Roll Number, otherwise it shall be deemed as an act of indulging in unfair means and action shall be taken as per rules.
2. Section-A contains Q-1 and Q-2 with Five (5) questions each. All the Ten very short type questions are compulsory and carry equal marks ($10 \times 2 = 20$).
3. Section-B contains Q-3, Q-4 and Q-5. Answer any TWO questions among the three descriptive type questions and carry 10 marks each. ($2 \times 10 = 20$).
4. Section-C contains Q-6 and Q-7 with FIVE (5) and FOUR (4) sub divisions. Answer any FOUR (4) and THREE (3) questions of choice among the short answer type questions and all carry 5 marks each. ($7 \times 5 = 35$)

Duration: 3 Hours

No. of Pages in this Question Paper: 1

Total Marks: 75

Section – A

1. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Define atrophy.
 - b) Define inflammation.
 - c) What is anemia?
 - d) What is angina pectoris?
 - e) Name any three mediator of inflammation.
2. Answer the following questions in very short (Maximum 20 words): 5 x 2 = 10
 - a) Name the enzyme, level of which elevated in Myocardial infarction.
 - b) What is significance of Creatinine level in blood?
 - c) Which enzymes level increases in liver injury? Name any two.
 - d) What is COPD?
 - e) Which neurotransmitter deficiency causes Parkinson's disease?

Section – B

Answer (any Two) of following questions in Long answer:

2 x 10 = 20

3. Discuss in detail about hypoxic cellular injury.
4. Classify hypertension, discuss etiopathogenesis of hypertension.
5. Classify diabetes and discuss etiopathogenesis of diabetes.

Section – C

6. Answer (any Four) of the following questions in short answer: 4 x 5 = 20
 - a) Write a short note on Rheumatoid arthritis.
 - b) Etiology of peptic ulcer.
 - c) Write note on pathogenesis of renal failure.
 - d) Note on mediators of inflammation.
 - e) Write note on Hypertrophy.
7. Answer (any Three) of the following questions in short answer: 3 x 5 = 15
 - a) Write note on Atherosclerosis.
 - b) Discuss pathogenesis of Alzheimer disease.
 - c) Discuss classification of cancer.
 - d) Discuss pathogenesis of tuberculosis.